Lab3-Score Distillation Sampling Implementation

Your Name (Student ID)
NYCU
your.email@example.com

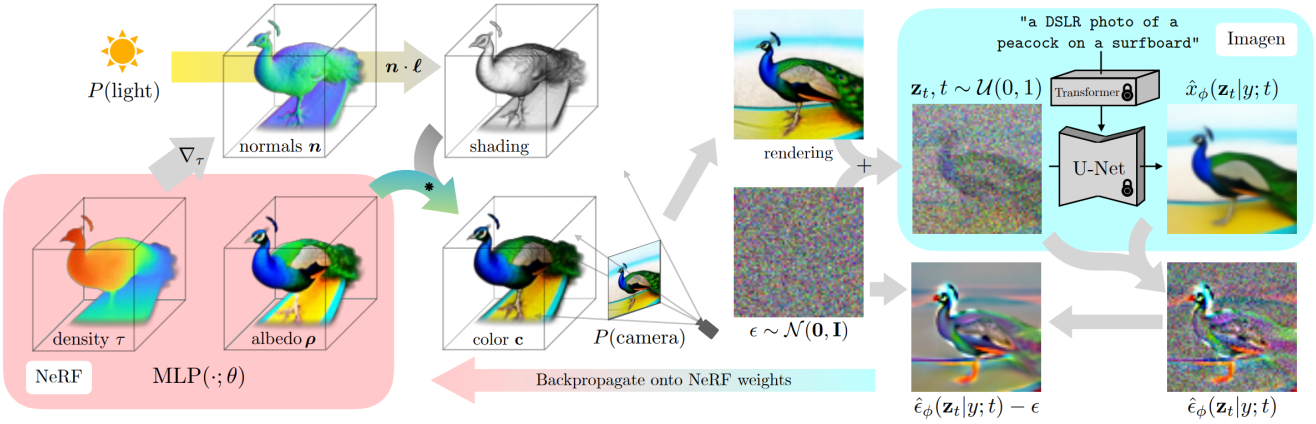


Figure 1. **Overview of score distillation.** Describe it briefly; cite its source if not yours.

1. Introduction

What problem does score distillation address, and how is it different from ordinary diffusion sampling? Keep it to a few sentences; you may refer to Fig. 1.

State the goal of this assignment: which methods you implemented, what you generated, and how you evaluated the results.

(Optional) Briefly outline the rest of the report.

2. Methodology

Introduce the background and notation shared by the three methods (*e.g.* the pretrained diffusion model [6], the forward noising process [2], classifier-free guidance [1]), so the following subsections can refer to them.

2.1. Score Distillation Sampling (SDS)

Concept. In your own words: what is being optimized in SDS [5], where does the gradient come from, and why is it a reasonable objective? Point out the approximation that makes SDS practical.

Implementation. Paste your `get_sds_loss()` into the code block below, then walk through it: the main steps, the design choices you made and why, and anything that was tricky to get right.

```
def get_sds_loss(
    self, latents, text_embeddings,
    guidance_scale=7.5,
):
    # Paste your implementation here
    ...
```

2.2. Variational Score Distillation (VSD)

Concept. Which weakness of SDS does VSD [8] try to fix? What extra component does it introduce (*cf*. LoRA [3]), and what is that component trained to do?

Implementation. Paste your `get_vsd_loss()` into the code block below, then walk through it: how the extra component is set up and updated, how its loss and the distillation loss interact, and which parts receive gradients.

```
def get_vsd_loss(
    self, latents, text_embeddings,
    guidance_scale=7.5, lora_loss_weight=1.0,
):
    # Paste your implementation here
    ...
```

2.3. Score Distillation via Inversion (SDI)

Concept. Which weakness of SDS does SDI [4] try to fix? How does it obtain the noisy sample differently (*cf*.

Table 1. **CLIP score** (↑, ViT-B/32) per prompt.

Prompt	SDS	VSD	SDI
A red bus driving on a desert road	0.0000	0.0000	0.0000
a boat in a river	0.0000	0.0000	0.0000
A cabin surrounded by forests	0.0000	0.0000	0.0000
A church beside a lake	0.0000	0.0000	0.0000
A villa close to the pool	0.0000	0.0000	0.0000
A castle next to a river	0.0000	0.0000	0.0000
A burger on the table	0.0000	0.0000	0.0000
A dog sitting on grass	0.0000	0.0000	0.0000
a cat sitting on a table	0.0000	0.0000	0.0000
A car on the road	0.0000	0.0000	0.0000
Average	0.0000	0.0000	0.0000

DDIM [7]), and how does its timestep schedule differ from SDS?

Implementation. Paste your `invert_noise()` (DDIM inversion) and `get_sdi_loss()` into the two code blocks below, then walk through them: how your inversion loop works, how the timestep changes over the iterations, and how and when the target is updated.

```
@torch.no_grad()
def invert_noise(
    self, latents, target_t, text_embeddings,
    guidance_scale=-7.5, n_steps=10, eta=0.3,
):
    # Paste your DDIM inversion here
    ...
```

```
def get_sdi_loss(
    self, latents, text_embeddings,
    guidance_scale=7.5,
    current_iter=0, total_iters=500,
    inversion_guidance_scale=-7.5,
    inversion_n_steps=10, inversion_eta=0.3,
    update_interval=25,
):
    # Paste your implementation here
    ...
```

3. Experiments & Results

3.1. Quantitative Results

Which method has the best average CLIP score (Tab. 1)? Are there prompts where the ranking changes, and why might that be? Does the score agree with your visual impression?

3.2. Qualitative Results

Compare the methods visually and point to concrete examples in Fig. 2. What are the typical strengths and failure cases of each method?

3.3. Ablation Study

Briefly describe the setup: which prompts and values were tested; all other hyperparameters follow Tab. 3. Table 2 lists the CLIP scores of all three ablations.

Table 2. **Ablation study.** Average CLIP score (↑).

(a) SDS: `guidance_scale`					
Value	7.5	15	25	50	100
CLIP	0.0000	0.0000	0.0000	0.0000	0.0000
(b) SDI: `inversion_n_steps`					
Value	1	3	5	10	25
CLIP	0.0000	0.0000	0.0000	0.0000	0.0000
(c) SDI: `sdi_update_interval`					
Value	25	75	200	500	1000
CLIP	0.0000	0.0000	0.0000	0.0000	0.0000

Table 3. **Hyperparameters** for each task.

Hyperparameter	SDS	VSD	SDI
`guidance_scale`	25	7.5	7.5
`lr`	0.01	0.01	0.005
`steps`	500	500	1000
`lora_lr`	–	1e-4	–
`lora_rank`	–	4	–
`lora_loss_weight`	–	1.0	–
`inversion_n_steps`	–	–	10
`inversion_guidance_scale`	–	–	-7.5
`inversion_eta`	–	–	0.3
`sdi_update_interval`	–	–	25

3.3.1. SDS: `guidance_scale`

How do the results change as the guidance scale increases (Fig. 3, Tab. 2(a)), and why? Which value works best, and how does it compare with the guidance scale typically used for ordinary sampling?

3.3.2. SDI: `inversion_n_steps`

How do the results and the runtime change with the number of inversion steps (Fig. 4, Tab. 2(b)), and why? Is there a trade-off?

3.3.3. SDI: `sdi_update_interval`

How do the results change with the update interval (Fig. 5, Tab. 2(c)), and why? What happens at the two extremes?

3.4. Hyperparameters

Fill in Tab. 3 with the settings you actually used (defaults are pre-filled from `main.py` / README) and justify any value you changed.

4. Discussion

4.1. Why the Original Diffusion Loss Fails

Explain how you produced the result in Fig. 6. Then explain intuitively (at most one equation) why directly optimizing the diffusion loss does not give realistic images, and connect your explanation to what is visible in the figure.



Figure 2. **Qualitative comparison** of SDS, VSD and SDI on the 10 evaluation prompts.

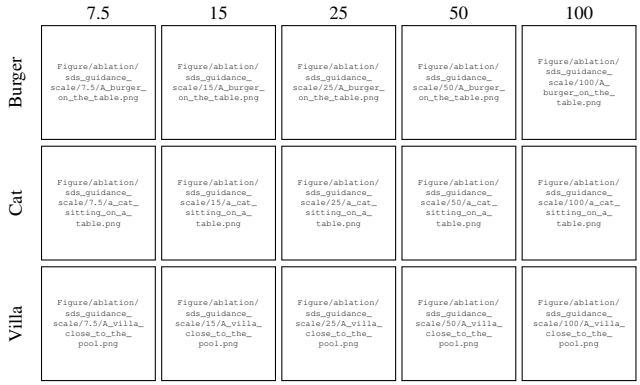


Figure 3. **Ablation on SDS `guidance_scale`.**

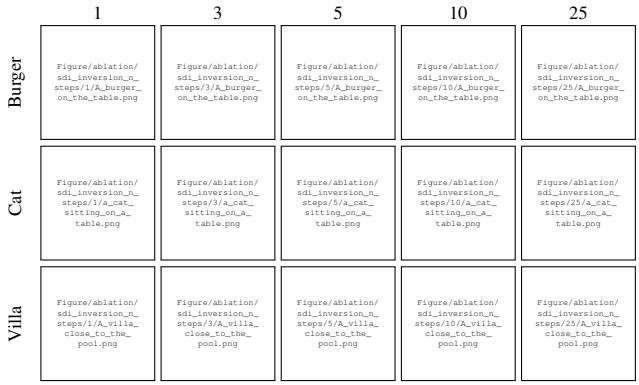


Figure 4. **Ablation on SDI `inversion_n_steps`.**

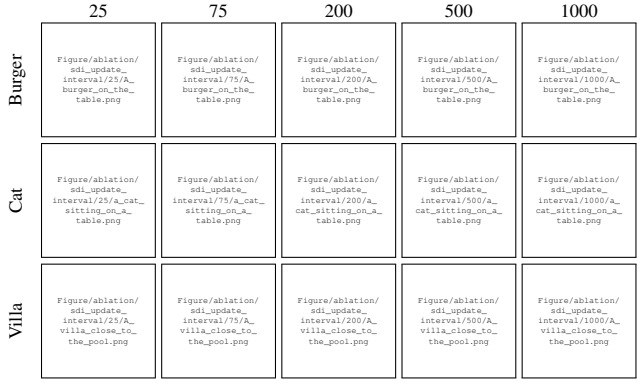


Figure 5. **Ablation on SDI `sdi_update_interval`.**

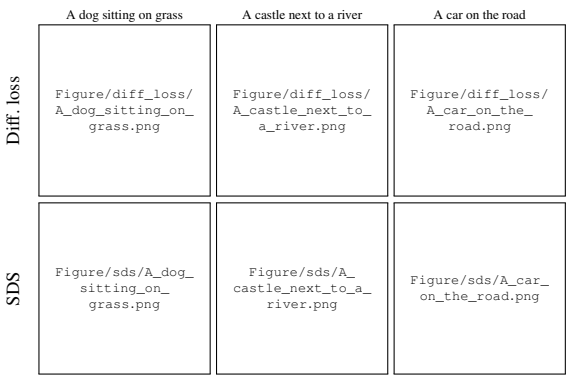


Figure 6. **Original diffusion loss** (top) vs. SDS (bottom).

4.2. Why SDI and VSD Outperform SDS

SDS. What are the typical artifacts of SDS in your results, and what in the SDS update causes them?
VSD. What does VSD change compared with SDS, and why does that lead to better results? Explain intuitively, without heavy math, and support it with your own figures or numbers.
SDI. What does SDI change compared with SDS, and why does that lead to better results? Explain intuitively, without heavy math, and support it with your own figures or numbers.

5. Conclusion

Summarize what you implemented, the main quantitative and qualitative findings, the key lessons from the ablations, and your answer to why VSD and SDI outperform SDS.

References

[1] Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. *arXiv preprint arXiv:2207.12598*, 2022. 1

[2] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Adv. Neural Inform. Process. Syst.*, 2020. 1

[3] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *Int. Conf. Learn. Represent.*, 2022. 1

[4] Artem Lukoianov, Haitz Sáez de Ocáriz Borde, Kristjan Greenewald, Vitor Campagnolo Guizilini, Timur Bagautdinov, Vincent Sitzmann, and Justin Solomon. Score distillation via reparametrized DDIM. In *Adv. Neural Inform. Process. Syst.*, 2024. 1

[5] Ben Poole, Ajay Jain, Jonathan T. Barron, and Ben Mildenhall. Dreamfusion: Text-to-3d using 2d diffusion. In *Int. Conf. Learn. Represent.*, 2023. 1

[6] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE Conf. Comput. Vis. Pattern Recog.*, 2022. 1

[7] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *Int. Conf. Learn. Represent.*, 2021. 2

[8] Zhengyi Wang, Cheng Lu, Yikai Wang, Fan Bao, Chongxuan Li, Hang Su, and Jun Zhu. Prolificdreamer: High-fidelity and diverse text-to-3d generation with variational score distillation. In *Adv. Neural Inform. Process. Syst.*, 2023. 1